

Proposed Solution

3. Project Design Phase | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
Credit Card Approval Prediction	Ankitha Vana	Finance	July 1, 2026

Solution Summary

The proposed solution is a machine learning based credit card approval prediction system. It processes applicant profile information and historical credit behavior, trains multiple classification models, selects the best model, and exposes it through a Flask web application.

The model does not replace human approval policies. Instead, it automates the first-level screening and helps analysts focus on applications requiring deeper review.

Core Solution Components

Component	Role
Data ingestion	Load applicant and credit status CSV files into Pandas DataFrames.
Data preprocessing	Clean duplicates, handle missing values, merge data, encode categories, and scale/transform fields as needed.
Feature engineering	Convert multi-class payment status records into binary risk labels for model learning.
Model training	Train Logistic Regression, Decision Tree, Random Forest, and XGBoost models.
Model selection	Compare models and save the best-performing classifier.
Flask app	Collect user inputs and display approval/rejection prediction.
Cloud deployment	Deploy the saved model through IBM Watson Machine Learning for scalable scoring.

Prediction Logic

The application receives applicant details, applies the same preprocessing steps used during training, passes the transformed feature vector to the saved model, and returns a binary decision: Approved or Rejected.